Customer Shopping Behavior Analysis

Uncovering insights from **3,600 transactions** across product categories 4 spending patterns, customer segments, and subscription behavior to guide strategic business decisions.

Datasst at a Clancs

3,900

Transactions

Total purchase records analyzed

18

Columns

Features spanning demographics, purchases, and behavior

37

Missing Values

In Review Rating column 4 imputed by category median

Demographics

Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Color

Shopping Behavior

Discounts, Promo Codes, Frequency, Reviews, Shipping

Exploratory Data Analysis in Python

→ Data Loading & Exploration

Imported dataset via pandas; used `df.info()` and `.describe()` for structure and summary stats.

→ Missing Data Handling

Imputed missing Review Rating values using the **median rating per product category**.

→ Feature Engineering

Created `age_group` by binning ages; derived `purchase_frequency_days` from purchase data.

→ Standardization & Database Load

Renamed columns to snake_case, dropped redundant `promo_code_used`, and loaded cleaned data into **PostgreSQL**.

SQL Analysis 4 Revenue & Discounts

Revenue by Cender

Compared total revenue generated by male vs. female customers.

High-Spending Discount Users

Identified customers who used discounts but still spent **above the average purchase amount**.

Discount-Dependent Products

Found the **5 products** with the highest percentage of discounted purchases.

Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

SQL Analysis 4 Products & Subscribers

Top 5 Products by Rating

Found products with the highest average review ratings.

Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

Repeat Buyers & Subscriptions

Checked whether customers with **>5 puFchases** are more likely to subscribe.

Top 3 Products per Category

Listed the most purchased products within each category.

Customer Segmentation

Segmentation Logic

Customers were classified into three tiers based on purchase history, enabling targeted engagement strategies.

- Revenue contribution was also calculated by **age group** to identify the highest-value demographics.

Clear all slicers

Customer analysis dashboard

gender

Female

Male

subscription_status

No

Yes

category

Accessories

Clothing

Footwear

shipping_type

- 2-Day Shipping
- Express
- Free Shipping
- Next Day Air
- Standard
- Store Pickup

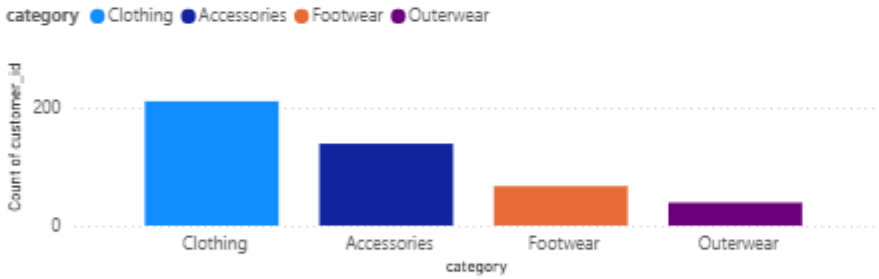
Percentage of Subscribers

456
no of customers

$60.57
average purcha...

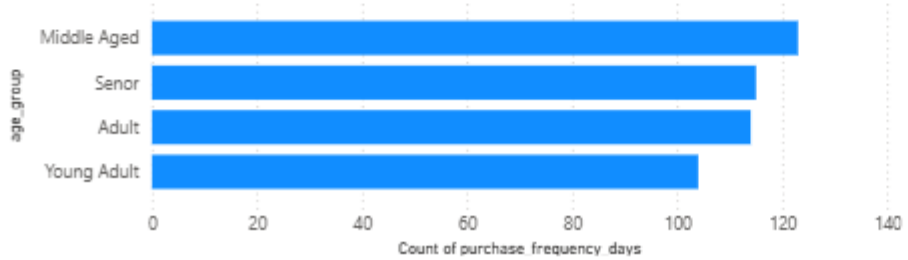
3.79
avg review rating

Sales by category

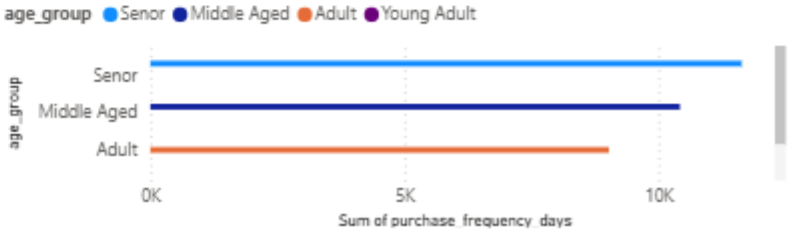


Revenue by category

Sales by age group



Revenue by age group



Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert non-subscribers.

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.

Product Positioning

Highlight top-rated and best-selling products in campaigns.